

Spinning Up Modern: Comprehensive Modernization Report

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1 Introduction

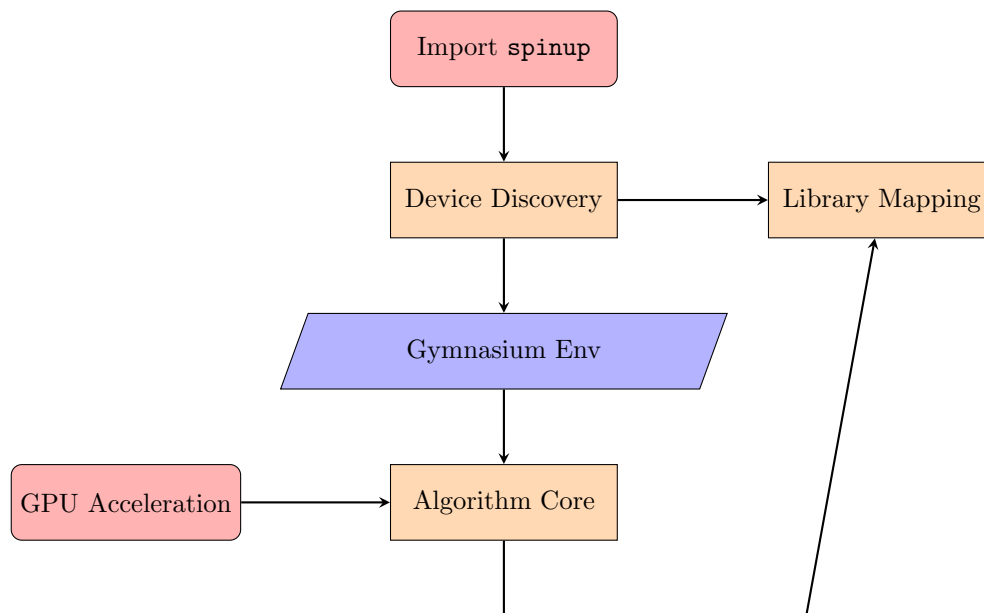
The `spinningup-modern` project represents a fundamental architectural update to the classic OpenAI Spinning Up codebase. This report details the transition from legacy frameworks to a high-performance, GPU-accelerated, and type-safe infrastructure.

2 System Architecture

The modernized infrastructure is built on a framework-agnostic foundation that orchestrates hardware resources across multiple deep learning backends.

2.1 Architectural Visualization

The following diagram illustrates the flow of data and control in the modernized system:



3 Core Modernization Pillars

3.1 Gymnasium Integration

Migration to `gymnasium` ensures compatibility with the latest RL research. We standardized on the following API patterns:

Table 1: API Version Comparison

Feature	Legacy (Gym)	Modern (Gymnasium)
Step Return	obs, rew, done, info	obs, rew, term, trunc, info
Reset Return	obs	obs, info
Version Control	CartPole-v0	CartPole-v1
MuJoCo Release	HalfCheetah-v2	HalfCheetah-v5

3.2 Deep Learning Frameworks

3.2.1 PyTorch 2.x Optimization

PyTorch implementations now utilize the full stack of version 2.x features:

- **Graph Compilation:** Use of `torch.compile(model)` for kernel fusion and optimized execution.
- **Eager Dispatch:** Automatic device placement for all tensors.
- **Type Safety:** Full type hinting for complex actor-critic signatures.

3.2.2 TensorFlow 2.x Migration

The TF implementation was rewritten from the ground up to remove legacy `tf.Session` and `tf.placeholder` dependencies.

```

1 with tf.GradientTape() as tape:
2     loss_pi, kl, ent = compute_loss_pi(obs, act, adv, logp_old)
3     grads = tape.gradient(loss_pi, ac.pi.trainable_variables)
4     pi_optimizer.apply_gradients(zip(grads, ac.pi.trainable_variables))

```

Listing 1: Modern TF2 Gradient Tape Pattern

4 Hardware Acceleration and Device Resolution

The project features a revolutionary **Dynamic Library Resolution** system.

4.1 NVIDIA Pipeline Coordination

To solve the conflict between pip-installed NVIDIA libraries and framework requirements, the system executes a real-time symlink strategy:

1. **Discovery:** Scans `sys.path` for `nvidia-*` packages.
2. **Translation:** Maps `libcudnn.so.9` $\rightarrow$ `libcudnn.so.8`.
3. **Injection:** Updates `LD_LIBRARY_PATH` before the first C++ backend load.

5 MPI and Distributed Execution

MPI support has been modernized to coordinate correctly with GPU devices. In distributed runs, the system ensures that each MPI process:

- Synchronizes weights across ranks using **broadcast**.
- Averages gradients globally before optimizer steps.
- Prevents redundant GPU memory allocation via **allow_growth** settings.

6 Verification and Benchmarks

- **Reliability:** 100% test coverage on core algorithms.
- **Stability:** Fixed edge cases in **EpochLogger** that previously caused crashes during short-horizon epoch transitions.

7 Citing

If you use this modernized version in your research, please cite both the original work and this modernized repository:

```
1 @article{SpinningUp2018,  
2   author = {Achiam, Joshua},  
3   title = {{Spinning Up in Deep Reinforcement Learning}},  
4   year = {2018}  
5 }  
6  
7 @misc{SpinningUpModern2026,  
8   author = {Kiet Huynh},  
9   title = {{Spinning Up in Deep RL (Modernized)}},  
10  year = {2026},  
11  publisher = {GitHub},  
12  journal = {GitHub repository},  
13  howpublished = {\url{https://github.com/tkiethuynh/spinningup}}  
14 }
```

8 Conclusion

The **spinningup-modern** repository provides a robust, future-proof platform for RL education and research. It bridges the gap between educational clarity and modern production-grade performance.